coping with all the hazards that life throws at their survival machines, and the judge is the ruthless judge of the court of survival. We shall come later to ways in which gene survival can be fostered by what appears to be altruistic behaviour. But the obvious first priorities of a survival machine, and of the brain that takes the decisions for it, are individual survival and reproduction. All the genes in the 'colony' would agree about these priorities. Animals therefore go to elaborate lengths to find and catch food; to avoid being caught and eaten themselves; to avoid disease and accident; to protect themselves from unfavourable climatic conditions; to find members of the opposite sex and persuade them to mate; and to confer on their children advantages similar to those they enjoy themselves. I shall not give examples—if you want one just look carefully at the next wild animal that you see. But I do want to mention one particular kind of behaviour because we shall need to refer to it again when we come to speak of altruism and selfishness. This is the behaviour that can be broadly labelled *communication*.*

A survival machine may be said to have communicated with another one when it influences its behaviour or the state of its nervous system. This is not a definition I should like to have to defend for very long, but it is good enough for present purposes. By influence I mean direct causal influence. Examples of communication are numerous: song in birds, frogs, and crickets; tail-wagging and hackle-raising in dogs; 'grinning' in chimpanzees; human gestures and language. A great number of survival-machine actions promote their genes' welfare indirectly by influencing the behaviour of other survival machines. Animals go to great lengths to make this communication effective. The songs of birds enchant and mystify successive generations of men. I have already referred to the even more elaborate and mysterious song of the humpback whale, with its prodigious range, its frequencies spanning the whole of human hearing from subsonic rumblings to ultrasonic squeaks. Mole-crickets amplify their song to stentorian loudness by singing down in a burrow which they carefully dig in the shape of a double exponential horn, or megaphone. Bees dance in the dark to give other bees accurate information about the direction and distance of food, a feat of communication rivalled only by human language itself.

The traditional story of ethologists is that communication signals evolve for the mutual benefit of both sender and recipient. For instance, baby chicks influence their mother's behaviour by giving high piercing